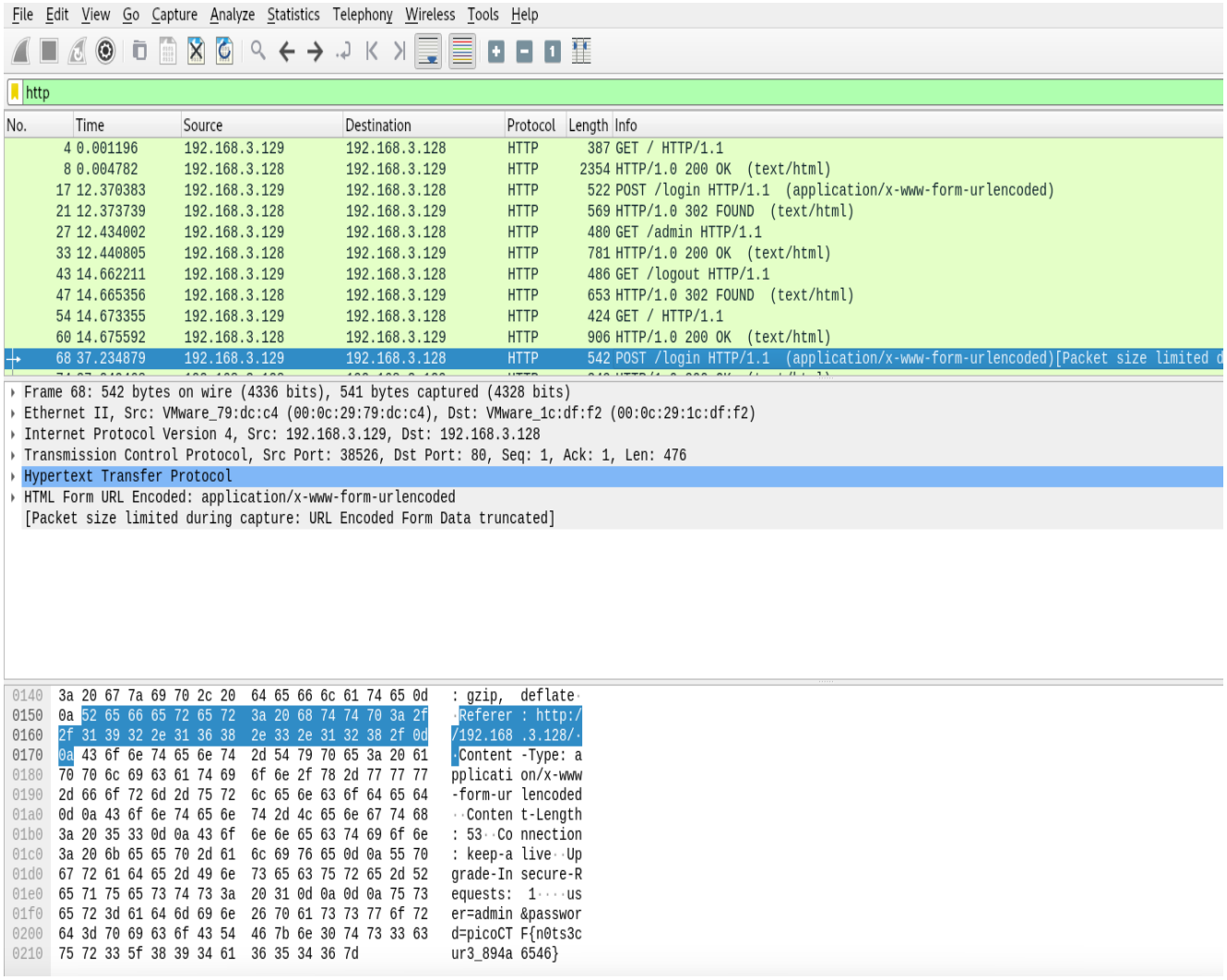
Writeups

-Azr43lKn1ght

<1> HTML.pcap

As the name of the file is a hint. We can search through http to find the flag.



Fl4g - picoCTF{n0ts3cur3_894a6546}

<2> DNS.pcap

No.	Time	Source	Destination	Protocol	Length	Info
43	1418.341495	192.168.2.1	192.168.2.12	DNS	87	Standard query 0xaaa0 ANY thisismyname.com OPT
55	1418.342859	192.168.2.12	192.168.2.1	DNS	316	Standard query response 0xaaa0 ANY thisismyname.com

```
▸ Frame 55: 316 bytes on wire (2528 bits), 316 bytes captured (2528 bits)
▸ Ethernet II, Src: PcsCompu_66:da:17 (08:00:27:66:da:17), Dst: 0a:00:27:00:00:16 (0a:00:27:00:00:16)
▸ Internet Protocol Version 4, Src: 192.168.2.12, Dst: 192.168.2.1
▸ User Datagram Protocol, Src Port: 53, Dst Port: 62905
▸ Domain Name System (response)
```

```
0060  00 01 00 00 01 2c 00 09  06 6d 79 6e 61 6d 65 c0   ·····,·· ·myname·
0070  19 c0 0c 00 0f 00 01 00  00 01 2c 00 04 00 05 c0   ........ ··,·····
0080  3e c0 0c 00 0f 00 01 00  00 01 2c 00 08 00 0a 03   >····· ··,·····
0090  6d 78 32 c0 3e c0 0c 00  0f 00 01 00 00 01 2c 00   mx2·>··· ·····,·
00a0  08 00 14 03 6d 78 33 c0  3e c0 0c 00 02 00 01 00   ····mx3· >·····
00b0  01 51 80 00 06 03 6e 73  31 c0 3e c0 0c 00 02 00   ·Q····ns 1·>·····
00c0  01 00 01 51 80 00 06 03  6e 73 32 c0 3e c0 0c 00   ···Q···· ns2·>···
00d0  10 00 01 00 00 01 2c 00  37 36 70 69 63 6f 43 54   ·····,·· 76picoCT
00e0  46 7b 77 34 6c 74 33 72  5f 77 68 31 74 33 5f 32   F{w4lt3r _wh1t3_2
00f0  64 36 64 33 63 36 63 37  35 61 61 33 62 65 37 66   d6d3c6c7 5aa3be7f
0100  34 32 64 65 62 65 64 38  61 64 31 36 65 33 62 7d   42debed8 ad16e3b}
0110  c0 0c 00 06 00 01 00 01  51 80 00 20 03 6e 73 31   ........ Q·· ·ns1
0120  c0 0c 03 64 6e 73 c0 0c  5b 41 6e 3e 00 00 0e 10   ···dns·· [An>····
0130  00 00 2a 30 00 01 51 80  00 00 0e 10               ··*0··Q· ····
```

Fl4g- picoCTF{w4lt3r_wh1t3_2d6d3c6c75aa3be7f42debed8ad16e3b}

<3>Trace.pcap

Mostly ICMP is the one we trace and the protocol hierarchy statistics is evident that there are data in it.

Once gone through the packets from ICMP , you can find file signature header of pkzip and chunks down the packets.

No.	Time	Source	Destination	Protocol	Length	Info
1217	4.122694323	192.168.1.200	185.245.99.2	ICMP	90	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
1222	4.159660793	192.168.1.200	185.245.99.2	ICMP	90	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
1233	4.207659549	192.168.1.200	185.245.99.2	ICMP	90	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 1247)
1244	4.255886878	192.168.1.200	185.245.99.2	ICMP	90	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 1259)
1247	4.293778802	185.245.99.2	192.168.1.200	ICMP	90	Echo (ping) reply id=0x0000, seq=0/0, ttl=52 (request in 1233)
1250	4.303010331	192.168.1.200	185.245.99.2	ICMP	90	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
1259	4.326376266	185.245.99.2	192.168.1.200	ICMP	90	Echo (ping) reply id=0x0000, seq=0/0, ttl=52 (request in 1244)
1264	4.353469210	192.168.1.200	185.245.99.2	ICMP	90	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 1282)
1271	4.392668732	192.168.1.200	185.245.99.2	ICMP	90	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 1290)
1282	4.427085834	185.245.99.2	192.168.1.200	ICMP	90	Echo (ping) reply id=0x0000, seq=0/0, ttl=52 (request in 1264)
1289	4.449825902	192.168.1.200	185.245.99.2	ICMP	90	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)

```
▸ Frame 1217: 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface eth0, id 0
▸ Ethernet II, Src: de:ad:00:00:be:ef (de:ad:00:00:be:ef), Dst: ca:fe:00:00:ba:be (ca:fe:00:00:ba:be)
▸ Internet Protocol Version 4, Src: 192.168.1.200, Dst: 185.245.99.2
▸ Internet Control Message Protocol
```

```
0000  ca fe 00 00 ba be de ad  00 00 be ef 08 00 45 00   ........ ......E·
0010  00 4c 00 01 00 00 40 01  9b 48 c0 a8 01 c8 b9 f5   ·L····@· ·H......
0020  63 02 08 00 10 b4 00 00  00 00 50 4b 03 04 14 00   c····· ··PK····
0030  09 00 08 00 92 ac 88 51  e2 f5 38 a1 6d 70 03 00   ·......Q ··8·mp··
0040  94 72 03 00 08 00 1c 00  66 6c 61 67 2e 6a 70 67   ·r····· flag.jpg
0050  55 54 09 00 03 d3 e3 cf  5f e7                     UT····· _·
```

So to extract the file , I made the following script.

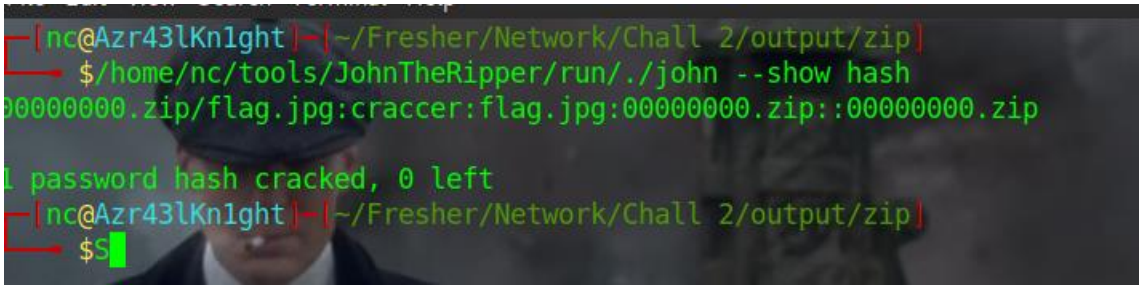
“

```
from scapy.all import *
f=rdpcap("Trace.pcap")
a=""
b=""
c=bytearray(len(f)*100)
d=0
for i in f:
  if ICMP in i and IP in i:
    if i[IP].src == "192.168.1.200":
      a=bytes(i)
      b=bytearray(a[42:])
      for j in range(len(b)):
        c[d]=b[j]
        d=d+1
open("trace.zip","wb").write(c)
```

“

Used foremost on the zip file to remove extra bytes in the end.

Used john the ripper to crack the zip file's password.



The file has the image flag.

<4>Buzz.pcap

When we look at the protocol hierarchy statistics we can see 8 packets of ICMP with lot of data.

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
Frame	100.0	1093	100.0	263363	10k	0	0	0
Ethernet	100.0	1093	5.8	15302	593	0	0	0
Internet Protocol Version 4	98.7	1079	8.2	21596	838	0	0	0
User Datagram Protocol	11.2	122	0.4	976	37	0	0	0
Network Time Protocol	2.5	27	0.5	1296	50	27	1296	50
Multicast Domain Name System	0.8	9	0.2	549	21	9	549	21
Domain Name System	7.9	86	2.2	5902	229	86	5902	229
Transmission Control Protocol	86.5	945	78.0	205390	7,970	602	71806	2,786
Transport Layer Security	33.2	363	76.2	200689	7,788	338	159277	6,181
Malformed Packet	0.3	3	0.0	0	0	3	0	0
Hypertext Transfer Protocol	0.2	2	0.5	1227	47	0	0	0
Online Certificate Status Protocol	0.2	2	0.2	554	21	2	554	21
Internet Group Management Protocol	0.4	4	0.0	64	2	4	64	2
Internet Control Message Protocol	0.7	8	4.5	11896	461	2	1112	43
Domain Name System	0.5	6	0.1	352	13	6	352	13
Address Resolution Protocol	1.3	14	0.1	392	15	14	392	15

On analysis of the data, we can find chunks of pk zip.

16	17.354534	10.30.8.102	192.168.42.83	ICMP	3528	Destination unreachable (Port unreachable)
19	17.442961	192.168.42.83	192.168.42.129	ICMP	160	Destination unreachable (Port unreachable)
444	91.835344	192.168.42.83	8.8.8.8	ICMP	131	Destination unreachable (Port unreachable)
449	92.302831	10.30.8.102	192.168.42.83	ICMP	3526	Destination unreachable (Port unreachable)
535	104.591591	10.101.236.32	192.168.42.83	ICMP	590	Destination unreachable (Fragmentation needed)
540	104.591594	10.101.236.32	192.168.42.83	ICMP	590	Destination unreachable (Fragmentation needed)
960	133.765764	192.168.42.83	8.8.8.8	ICMP	119	Destination unreachable (Port unreachable)
961	133.792972	10.30.8.102	192.168.42.83	ICMP	3524	Destination unreachable (Port unreachable)

```
▸ Frame 16: 3528 bytes on wire (28224 bits), 3528 bytes captured (28224 bits)
▸ Ethernet II, Src: ba:41:a7:b7:29:50 (ba:41:a7:b7:29:50), Dst: 1e:70:33:1e:96:03 (1e:70:33:1e:96:03)
▸ Internet Protocol Version 4, Src: 10.30.8.102, Dst: 192.168.42.83
▸ Internet Control Message Protocol
▸ Domain Name System (query)
```

```
0000  1e 70 33 1e 96 03 ba 41  a7 b7 29 50 08 00 45 28   ·p3····A ··)P··E(
0010  0d ba e8 9a 00 00 3d 01  8a 01 0a 1e 08 66 c0 a8   ......=· ·····f··
0020  2a 53 03 03 fa df 00 00  00 00 45 28 00 66 20 e0   *S····· ··E(·f ·
0030  40 00 3d 11 1f 00 c0 a8  2a 53 0a 1e 08 66 dc b4   @·=····· *S···f··
0040  00 35 00 52 5c c9 40 b9  01 00 00 01 00 00 00 00   ·5·R\·@· ........
0050  00 00 21 69 32 2d 71 62  71 76 76 65 61 6e 73 70   ··!i2-qb qvveansp
0060  6b 65 6a 6c 77 74 61 74  71 70 63 76 79 62 6c 62   kejlwtat qpcvyblb
0070  6e 64 67 6f 04 69 6e 69  74 0d 63 65 64 65 78 69   ndgo·ini t·cedexi
0080  73 2d 72 61 64 61 72 03  6e 65 74 00 00 01 35 30   s-radar· net···50
0090  34 42 30 33 30 34 31 34  30 33 30 30 30 30 30 38   4B030414 03000008
00a0  30 30 30 31 39 39 30 32  34 44 32 44 33 37 30 36   00019902 4D2D3706
00b0  44 35 36 31 31 33 30 30  30 30 30 43 31 41 30 30   D5611300 000C1A00
00c0  30 30 30 38 30 30 30 30  30 30 36 36 36 43 36 31   00080000 00666C61
00d0  36 37 32 45 37 30 36 45  36 37 44 44 35 38 36 37   672E706E 67DD5867
00e0  35 30 44 33 37 37 31 38  38 45 41 35 38 35 42 36   50D37718 8EA585B6
```

So made the following script to extract it.

“

```
from scapy.all import *

f=rdpcap("Buzz.pcap")

a=""

b=""

c=""

d=""

for i in f:

  if ICMP in i and IP in i:

    if i[IP].src=="10.30.8.102":

      a=bytes(i)

      b=str(a[142:])

      c=c+b[2:-1]

print(c)
```

“

Then made it into a zip file by xxd.

inctf{_s0meTim3s_u_h4v3_t0_lOOk_3v3ryWh3r3_cl0s3r_T0_G3T_th3_wh0l3!}

The flag is inside the zip file.

<5>Data.pcap

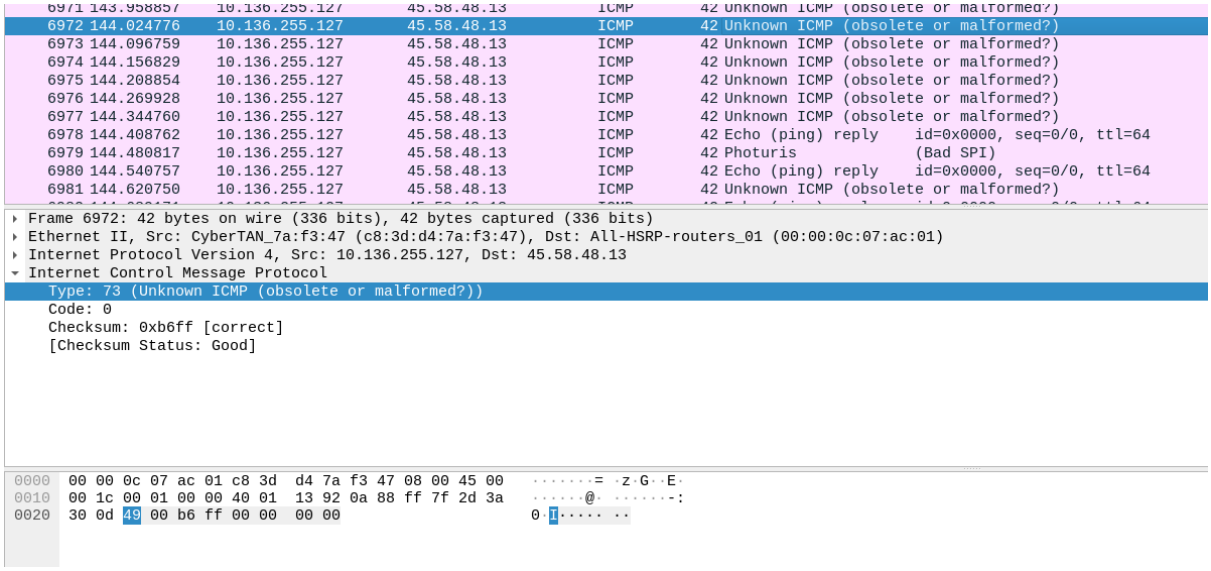
After protocol hierarchy, just went into ICMP data packets.

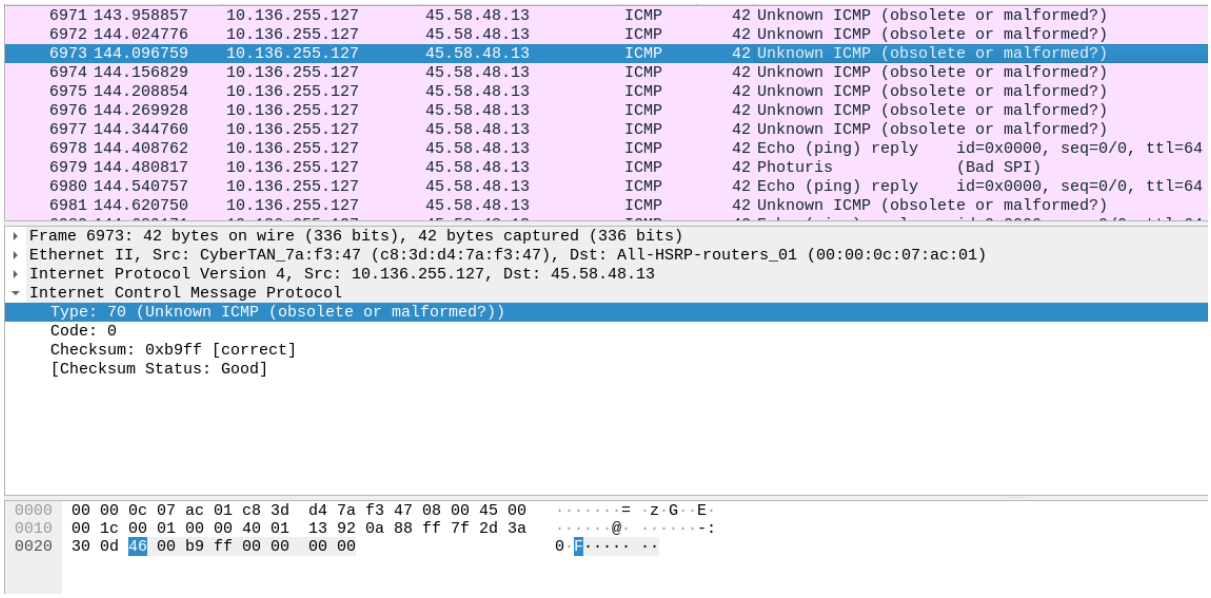
Was able to figure out header of GIF file in 3 different packets.

6971	143.958857	10.136.255.127	45.58.48.13	ICMP	42	Unknown ICMP (obsolete or malformed?)
6972	144.024776	10.136.255.127	45.58.48.13	ICMP	42	Unknown ICMP (obsolete or malformed?)
6973	144.096759	10.136.255.127	45.58.48.13	ICMP	42	Unknown ICMP (obsolete or malformed?)
6974	144.156829	10.136.255.127	45.58.48.13	ICMP	42	Unknown ICMP (obsolete or malformed?)
6975	144.208854	10.136.255.127	45.58.48.13	ICMP	42	Unknown ICMP (obsolete or malformed?)
6976	144.269928	10.136.255.127	45.58.48.13	ICMP	42	Unknown ICMP (obsolete or malformed?)
6977	144.344760	10.136.255.127	45.58.48.13	ICMP	42	Unknown ICMP (obsolete or malformed?)
6978	144.408762	10.136.255.127	45.58.48.13	ICMP	42	Echo (ping) reply id=0x0000, seq=0/0, ttl=64
6979	144.480817	10.136.255.127	45.58.48.13	ICMP	42	Photuris (Bad SPI)
6980	144.540757	10.136.255.127	45.58.48.13	ICMP	42	Echo (ping) reply id=0x0000, seq=0/0, ttl=64
6981	144.620750	10.136.255.127	45.58.48.13	ICMP	42	Unknown ICMP (obsolete or malformed?)

```
▸ Frame 6971: 42 bytes on wire (336 bits), 42 bytes captured (336 bits)
▸ Ethernet II, Src: CyberTAN_7a:f3:47 (c8:3d:d4:7a:f3:47), Dst: All-HSRP-routers_01 (00:00:0c:07:ac:01)
▸ Internet Protocol Version 4, Src: 10.136.255.127, Dst: 45.58.48.13
▾ Internet Control Message Protocol
    Type: 71 (Unknown ICMP (obsolete or malformed?))
    Code: 0
    Checksum: 0xb8ff [correct]
    [Checksum Status: Good]
```

```
0000  00 00 0c 07 ac 01 c8 3d  d4 7a f3 47 08 00 45 00   ·····= ·z·G··E·
0010  00 1c 00 01 00 00 40 01  13 92 0a 88 ff 7f 2d 3a   ····· @· ·····-:
0020  30 0d 47 00 b8 ff 00 00  00 00                     0·G····· ··
```





So I made a script to extract the data.

“

```
from scapy.all import *

f=rdpcap("Data.pcap")

st=""

flag=bytearray(10000)
```

```
a=0

for i in f:

  if ICMP in i and IP in i:

    if i[IP].src=="10.136.255.127":

      st=bytes(i)

      flag[a]=st[34]

      a=a+1

open("flag.gif","wb").write(flag)
```

“

Then we get the Flag!!

flag{we_ride_at_midnight}

Github: https://github.com/Azr43lKn1ght/Azr43lKn1ght